Separating DNA Fragments

Biology 2e chapter 17 reading connection | Teacher guide / classroom preparation

Learning target

How can migration distance help estimate DNA fragment size and compare samples?

Timing and materials

One to two class periods, depending on casting and staining method.

Gel chamber and power supply, casting tray and comb, agarose, buffer, prepared DNA fragments and ladder, loading dye, micropipette and tips, approved stain, heating equipment, ruler, and viewer if required by the stain.

Classroom route

1. Use prepared DNA and the supplier-recommended agarose percentage, buffer, voltage and stain. A visible stain avoids a fluorescence viewer. Teacher prepares hot agarose and allows it to set before use.

2. Load ladder, two references and an unknown in separate wells. Keep wells at the negative end so DNA travels toward the positive electrode. Close the lid before energizing. Never touch buffer or open the chamber with power connected.

3. Stop and disconnect power before staining or handling the gel. Measure distance from each well to its bands. Plot ladder distance against log10(base pairs) and interpolate unknown sizes within its calibrated range. Repeat if supplies allow.

Pilot the physical procedure with your own equipment. Follow school handling and disposal rules and the purchased kit instructions. The virtual numerical settings are not an equipment prescription.

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Hybrid or substitute assignment

Assign all virtual, all physical, or specified samples in each mode. The classroom entry button is available during collection. Students complete the same prediction and evidence questions. Record classroom time series on the worksheet; the online classroom form stores endpoint measurements and notes. Keep sources and measurement scales separate.

Guided virtual sequence

Students prepare each sample, inspect the apparatus or classify specimens, then explicitly collect data. All assigned trials are required before exploration. Reload pauses the clock and restores saved work. Downloads retain source labels; no automatic grade is submitted.

Expected reasoning

Unknown bands are modeled at 3000, 1500 and 700 bp, matching reference A. Accept reasonable interpolated estimates. Distance is approximately linear in log10(size) over the useful range. A matching classroom pattern does not establish a real-world personal identity.

Feedback and scoring

Criterion	Points	Evidence
Prediction and controls	0-2	Testable claim; comparable samples and controlled variables.
Observations	0-2	Complete, labeled records with units and source.
Interpretation	0-2	Correct calculations or categorical comparisons; claim tied to evidence.
Limitations	0-2	Specific uncertainty and a useful follow-up investigation.

Score reasoning and data quality, not agreement of physical results with the simulation. Manual tallies are student observations; a matching total does not prove every category is correct. Do not replace student counts with the model answer. The optional Sordaria extension retains classification feedback.

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Model limitations

Original band patterns and a simplified log-distance model; no supplier mystery or real forensic sample. Diffusion, DNA topology, overloading and nonlinear extremes are omitted. Illustrative voltage and timing are not a physical equipment protocol.

Troubleshooting

Confirm the preparation steps are finished before starting or collecting. Corn and fly activities require manual tallies summing to the sample size plus review; optional specimen marks do not classify or tally automatically. Chromosome models require all four gametes plus a reviewed tally; cell measurements require a focused 40x view; gels require staining and three size estimates; fluorescence labs require the viewer. Use the visible Collect data and continue button to advance.

For physical work, preserve unexpected observations, check the control and instrument setup, then repeat with fresh material where appropriate. Never replace measured results with the virtual answer.

Evidence and optional classroom resources

Classroom electrophoresis reference

https://www.bio-rad.com/webroot/web/pdf/lse/literature/4006096G.pdf

External resources remain on their publishers' websites. This packet contains original text and tables, not supplier worksheets. Chapter numbers are reading connections. No supplier or textbook publisher endorsement is implied.

Teacher pilot checklist

Confirm availability and preparation time; test controls; check units and observation windows; verify handling/disposal; choose class grouping and accessible alternatives; keep notes for revision.